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ORIGINAL ARTICLE

Long-term history and synchrony of mountain pine beetle outbreaks in lodgepole pine forests

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ABSTRACT

Aim A subcontinental-scale outbreak of mountain pine beetle (MPB; *Dendroctonus ponderosae*) has affected millions of hectares of forest in the western USA and Canada over the past 15 years. Little research has examined the long-term and broad-scale history of MPB outbreaks, which is necessary to provide a baseline for comparing current and future outbreak extents. We examine the long-term history of MPB outbreaks in western Colorado, and analyse the synchrony of previous outbreaks across western North America.

Location Western Colorado for tree-ring analysis; western North America for comparative analysis.

Methods We used dendroecological methods to reconstruct the history of MPB outbreaks since the 1700s in lodgepole pine (*Pinus contorta*) forests in Colorado. We then combined these new records with previously published data on MPB outbreaks to examine the historical synchrony of outbreaks across western North America.

Results The tree-ring record indicated that multiple cross-site MPB outbreaks occurred in Colorado lodgepole pine forests, initiating *c.* 1760s, 1780s, 1820s–1830s, 1860s, 1910s, 1960s and 1980s. Comparative analysis indicated that these outbreak dates coincide with documented and reconstructed MPB outbreaks in British Columbia, Colorado, Idaho, Montana, Wyoming and Utah.

Main conclusions Several subcontinental-scale MPB outbreaks have occurred in lodgepole pine forests over the past three centuries. Although the current study does not address the intensity of past outbreaks, it does indicate that previous outbreaks appear to have been highly synchronous across western North America. The subcontinental, rather than purely local, nature of these outbreaks is important for putting recent and future outbreaks of MPB into a broader ecological context. This study presents the longest tree-ring reconstruction of MPB outbreaks in the southern Rocky Mountains available to date, as well as the first subcontinental long-term comparative analysis of MPB outbreaks in western North America.

Keywords

Bark beetles, Colorado, *Dendroctonus ponderosae*, dendroecology, disturbance regime, forest disturbance, outbreak history, *Pinus contorta*, Rocky Mountains, subalpine forest.

INTRODUCTION

Shifts in temperature and precipitation associated with a changing climate are expected to affect coniferous forests (McNulty & Aber, 2001) and to increase the potential for complex shifts in forest disturbance regimes (Bentz *et al.*, 2010; Turner, 2010). A global shift in the extent and severity of forest insect outbreaks has been attributed to climate warming (Ayres & Lombardero, 2000; Logan *et al.*, 2003; Weed *et al.*, 2013). Consequently, many recent outbreaks are

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believed to lie outside the historical range of variation. A fundamental understanding of past disturbance dynamics is required to characterize such novel ecosystem trajectories (Lenton *et al.*, 2008). Although important advances have been made in understanding the effects of warming on the physiology of insects and hosts, the long-term dynamics of some important insect outbreaks remain poorly understood. A historical perspective is critical in determining whether, and in what ways, recent climatically driven outbreaks are outside their historical range of variation, rather than being infrequent but normal events. An ongoing outbreak of mountain pine beetle (MPB; *Dendroctonus ponderosae* Hopkins, 1902) in western North America has been notably synchronous, widespread and severe (Raffa *et al.*, 2008) and is widely believed to be unprecedented. Surprisingly, however, there is a scarcity of research on the long-term and broad-scale dynamics of MPB outbreaks that would be needed to put the ongoing outbreak into the context of the historical range of variation. Without such research, the assertion that recent outbreaks are unprecedented is largely speculative. Here, we present the longest reconstruction of MPB outbreaks in the southern Rocky Mountains available to date, as well as the first subcontinental comparative analysis of MPB outbreaks in western North America.

With climate change, forest disturbance regimes are expected to change in frequency, extent and magnitude (Safranyik *et al.*, 2010; Westerling *et al.*, 2011; Choat *et al.*, 2012; Weed *et al.*, 2013). For example, wildfire in high-elevation Rocky Mountains forests is strongly linked to dry climate conditions (Bessie & Johnson, 1995; Kipfmueller & Baker, 2000; Buechling & Baker, 2004; Heyerdahl *et al.*, 2008; Morgan *et al.*, 2008). With increasingly warm and dry conditions, wildfires in the western USA are occurring more frequently and over longer fire seasons (Westerling *et al.*, 2006). In the Greater Yellowstone Ecosystem in Wyoming, Westerling *et al.* (2011) predict that continued climate warming may push this system beyond an ecological threshold, which may transform the landscape from dense, conifer-dominated forest to sparse woodland and non-forest vegetation. Outbreaks of MPB are also linked to warmer and drier conditions. MPB winter survival, life-cycle timing and attack synchrony are directly affected by temperature (Bentz *et al.*, 2010; Logan *et al.*, 2010), and drier conditions indirectly increase the frequency of outbreaks by increasing host susceptibility (Safranyik, 2004). With recent climatic trends, outbreaks have expanded beyond their presumed historical range and are affecting forests at higher elevations and higher latitudes (Carroll *et al.*, 2004; Raffa *et al.*, 2013). In Alberta, for example, jack pine (*Pinus banksiana*) forests do not have a previous record of MPB infestation (Bentz *et al.*, 2010; Safranyik *et al.*, 2010) as the local climate has been characterized as being too cold to sustain outbreak conditions (Carroll *et al.*, 2004). Climate change now has resulted in more suitable summer (Logan & Powell, 2001) and winter (Carroll *et al.*, 2004) conditions for MPB populations, and these forests are being infested at the leading edge of the 2000s–2010s outbreak, exposing vast swathes of forest to future outbreak (Safranyik *et al.*, 2010; Cullingham *et al.*, 2011).

Large forest disturbances, such as high-severity fire and outbreaks of some insects, typically have relatively long return intervals and require historical reconstruction on the order of centuries to characterize their disturbance regimes. In the absence of such long-term data, incorrect conclusions may be drawn about whether large disturbance events are indeed unprecedented consequences of global environmental change. For example, high-severity fires burned more than 300,000 ha of forest in Yellowstone National Park in 1988, which was an order of magnitude larger than previously recorded fire years (Christensen *et al.*, 1989), sparking debate about the historical precedence of large fires. Historical reconstructions of fire using tree rings and lake sediments showed, however, that such large fires were within the historical range of variability in these ecosystems (Romme & Knight, 1981, 1982; Romme & Despain, 1989; Millspaugh & Whitlock, 1995). Similarly, the 2000s–2010s outbreak of MPB in western North America has often been suggested to be unprecedented, although the extent and synchrony of MPB outbreaks over the past centuries have not been thoroughly characterized. Instead, comparisons are typically made to outbreaks only over the past decades – a period that is very short given the long return intervals of other disturbances in the affected forests.

MPB is the leading cause of insect-related mortality in the pine-dominated forests of western North America (Schmid & Mata, 1996). In Colorado's subalpine zone (*c.* 2850–3500 m), the primary host for MPB is lodgepole pine (*Pinus contorta* Douglas ex Loudon). An outbreak of MPB can shift stand dominance to Engelmann spruce (*Picea engelmannii* Parry ex Engelm.) and subalpine fir [*Abies lasiocarpa* (Hook.) Nutt.] if there is adequate suppressed spruce and fir in the understory (Schmid & Amman, 1992). Endemic MPB populations primarily infest weakened and injured trees (Schmid & Amman, 1992). When suitable conditions are present and populations increase, MPB initiates a pheromone-mediated mass attack to overwhelm the defences of healthy, living trees (Raffa & Berryman, 1983). During attacks, the beetles construct larval galleries as they feed on the host's inner phloem layer, with the adults carving *c.* 30-cm J-shaped tunnels (galleries) that are easily visible under the bark (Safranyik & Wilson, 2006). As the beetles feed, they introduce blue-stain fungi (ophiostomatoid fungi), and the growth of these fungi impedes nutrient transport (Ballard *et al.*, 1984). During an outbreak, larger trees (typically > 10 cm in diameter) are preferentially attacked, but smaller-diameter trees can also be attacked under extreme conditions (Schmid & Mata, 1996). Following a bark beetle outbreak, non-host and surviving host tree species benefit from increased resources previously utilized by host trees and an increase in growth becomes visible in the annual rings of these surviving trees (Waring & Pitman, 1985; Veblen *et al.*, 1991a).

Several previous studies have documented or reconstructed past MPB outbreaks in the USA and Canada

Table 1 Approximate dates of canopy disturbance attributable to *Dendroctonus ponderosae* outbreak in the Independence Pass area of Colorado, USA, as inferred from this study and outbreak dates from previous studies. The first 10 rows correspond to sites from this study. The following rows correspond to previous studies (see Table 2 for details). Columns represent outbreak initiation dates.

Site	Outbreak dates from this study									
1			1760*			1860*	1910*		1965*	1985*
2				1780*	1810*		1910*	1945*	1965*	1985*
3		1710*	1760*			1860*	1910*		1965*	1985*
4			1760*	1790*			1910*		1965*	1985*
5	1660*			1785*			1910*		1965*	1985*
6			1760*		1830*		1910*		1965*	1985*
7				1785*	1835*		1910*		1965*	1985*
8	1680*			1785*	1830*	1860*	1910*		1965*	1985*
9				1785*	1825*	1860*	1910*	1940*	1965*	1985*
10			1760*	1780*			1910*		1965*	1985*
Previous study	Outbreak dates from other studies									
a										1980s*†
b								1940s–1950s*		1980s*
c										1980s*†
d					1830s*	1860s–1870s*	1900s–1910s*	1940s–1950s*†		1980s*†
e								1940s*		1980s–1990s*†
f										1970s*†
g						1850s–1870s*		1930s*	1950s–1960s*	1980s*†
h						1870s*	1890s–1910s*	1930s–1940s*	1960s*	1970s–1980s*†
i								1920s–1930s*†	1960s*	1970s*†
j										1970s*†
k										1970s–1980s*†
l							1900s–1920s†	1930s†	1950s–1960s†	1970s–1980s†
m			1760s*	1780s*		1860s*	1890s–1900s*	1920s–1940s*†		1970s–1980s*†
n										1980s†
o										1970s*†
p										1970s†
q							1890s*	1920s–1950s*	1960s*	1970s†
r				1780s†		1870s†	1910s†	1920s†		1980s*†
s							1900s–1910s†			

Outbreak dates from this study are inferred from a combination of mortality dates, quantified releases and cross-site synchrony.
*Reconstructed using dendrochronological methods.
†Documentary evidence.

(Tables 1 & 2). The majority of reconstructed outbreaks have been in British Columbia, towards the northern limit of MPB range. The earliest dendroecologically (tree-ring) reconstructed outbreaks in North America date to the late 18th century (central British Columbia; Hawkes *et al.*, 2003) and early 19th century (northern British Columbia; Hrinkevich & Lewis, 2011). A smaller number of studies have documented or reconstructed outbreaks in Colorado, Wyoming, Montana, Utah and Idaho, but most of these outbreaks have been limited to recent decades. The earliest dendroecologically reconstructed outbreaks in the southern Rocky Mountains occurred in the 1940s (Kulakowski & Jarvis, 2011; Kulakowski *et al.*, 2012), but historical records describe outbreaks in north-eastern Utah in the late 18th and late 19th centuries (Thorne, 1935). To our knowledge, no study has examined the potential synchrony of historical MPB outbreaks across these regions at a subcontinental scale, and so little is known about their broad-scale and long-term dynamics.

This study addresses the following research questions. What is the long-term history of MPB outbreaks in western Colorado? How synchronous have historical MPB outbreaks been throughout western North America? We use tree-ring methods to reconstruct the history of MPB outbreaks over the past four centuries in western Colorado, and we compare the synchrony of these outbreaks with those that have been previously documented or reconstructed across western North America. This study is the first to reconstruct MPB outbreaks in the southern Rocky Mountains prior to the 20th century and also the first to systematically assess subcontinental synchrony of historical MPB outbreaks.

MATERIALS AND METHODS

Study area

Sites were selected using methods similar to those used in previous reconstructions of MPB outbreaks (Axelson *et al.*,

Table 2 Details of the previous studies referred to in Table 1.

	Study	Location
a	Smith *et al.* (2012)	North-west Colorado, USA
b	Kulakowski *et al.* (2012)	North-west Colorado, USA
c	Schoennagel *et al.* (2012)	North-west Colorado, USA
d	Hrinkevich & Lewis (2011)	North-central British Columbia, Canada
e	Kulakowski & Jarvis (2011)	North-west Colorado/south-east Wyoming, USA
f	Simard *et al.* (2011)	Greater Yellowstone, Wyoming, USA
g	Axelson *et al.* (2010)	Central British Columbia, Canada
h	Alfaro *et al.* (2010)	British Columbia, Canada
i	Axelson *et al.* (2009)	Southern British Columbia, Canada
j	Sibold *et al.* (2007)	Rocky Mountain National Park, Colorado, USA
k	Campbell *et al.* (2007)	Central British Columbia, Canada
l	Taylor *et al.* (2006)	British Columbia, Canada
m	Hawkes *et al.* (2003)	Central British Columbia, Canada
n	Stone & Wolfe (1996)	Northern Utah, USA
o	Heath & Alfaro (1990)	British Columbia, Canada
p	Klein *et al.* (1979)	South-east Idaho, USA
q	Roe & Amman (1970)	North-west Wyoming, USA
r	Thorne (1935)	North-east Utah, USA
s	Evenden (1934)	North-west Montana, USA

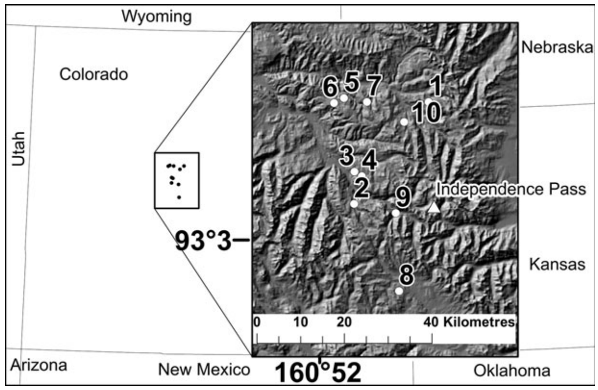

Figure 1 Locations of sampling sites in western Colorado (Albers equal-area projection).

2010; Hrinkevich & Lewis, 2011; Smith *et al.*, 2012). Old forests in Colorado with lodgepole pine dominant or codominant were identified using the GIS vegetation layer 'R2-Veg' (USDA Forest Service, Golden, CO, USA; available at: http://www.fs.fed.us/r2/gis/; accessed 1 April 2008). Within this extent of lodgepole pine, satellite imagery (Google Earth 5, Google, Mountain View, CA; http://earth.google.com/) at fine spatial resolution (*c.* 4–15 m) was used to identify potential sampling sites that included downed dead lodgepole pine orientated in random directions.

Twenty-five potential sites were scouted in the field, and sites were selected for sampling if they met the following criteria: (1) they did not contain evidence of post-stand-origin fire, in the form of fire scars on trees within or surrounding the site, visible presence of charcoal, or clearly delineated subcanopy cohorts that may have resulted from a low-severity fire; (2) they contained identifiable standing or fallen dead lodgepole pine trees lying in random directions and with similar levels of decay; (3) they were at least 1 km across the longest axis; and (4) dead lodgepole pine had clear MPB galleries and blue-stain fungus on core samples. Ten of the 25 sites met these criteria; they are located in a 2000-km^2 area of subalpine forest in western Colorado just west of Independence Pass (Fig. 1). They are a mix of sites dominated by lodgepole pine and those codominated by lodgepole pine, Engelmann spruce (*Picea engelmannii*) and subalpine fir (*Abies lasiocarpa*), and range in elevation from 2988 to 3345 m. The climate is continental, with monthly mean temperatures ranging from −7.7 °C in January to 15.9 °C in July, and monthly mean precipitation ranging from *c.* 300 mm in January to *c.* 400 mm in July (1963–2007, 2380 m, Station 055507; Western Region Climate Center, Reno, NV, USA; http://www.wrcc.dri.edu/). Historically, most lodgepole pine forests in the southern Rocky Mountains were initiated after stand-replacing fires (Whipple & Dix, 1979; Kipfmueller & Baker, 2000; Sibold *et al.*, 2006; Schoennagel *et al.*, 2007), and they were successionally replaced at some sites by Engelmann spruce and subalpine fir (Peet, 1981; Veblen & Lorenz, 1986).

Sampling and processing

MPB outbreaks were reconstructed using well-established dendroecological methods that have been used extensively in reconstructing MPB outbreaks (Campbell *et al.*, 2007; Alfaro *et al.*, 2010; Axelson *et al.*, 2010; Hrinkevich & Lewis, 2011; Kulakowski *et al.*, 2012; Schoennagel *et al.*, 2012; Smith *et al.*, 2012) and are similar to the methods used to reconstruct outbreaks of spruce beetle (Baker & Veblen, 1990; Veblen *et al.*, 1991a; Eisenhart & Veblen, 2000; Berg *et al.*, 2006; Kulakowski & Veblen, 2006; Sherriff *et al.*, 2011). Dendroecological detection of outbreaks involved determining the mortality dates of dead host trees and quantifying the increases in average ring widths in unaffected trees that were released from competition after the outbreak (Waring & Pitman, 1985; Veblen *et al.*, 1991a). Severe wind events or other disturbances could also have resulted in an increase in the average ring width of surviving trees by eliminating canopy cover and reducing competition (e.g. Veblen *et al.*, 1989, 1991b); mortality from MPB outbreaks could be distinguished from wind storms in the dendroecological record because outbreaks resulted in azimuths of fallen trees that were random or determined by site aspect, host-specific mortality and the presence of MPB galleries and blue-stain fungus. Fires in Colorado's subalpine forests have typically been stand-replacing with few surviving trees (Buechling & Baker, 2004; Sibold *et al.*, 2006; Schoennagel *et al.*, 2007) and so the dendroecological signal of fire is distinctly different from that of MPB outbreaks in these ecosystems.

Increment cores were collected at each sample site from at least 20 dead host (lodgepole pine) and at least two adjacent live host or non-host trees per each dead tree to aid in cross-dating years of mortality. Each dead lodgepole pine was inspected for the presence of blue-stain fungi in the core and characteristic J-shaped galleries under the remaining bark. Cores from dead lodgepole were only collected if the tree contained both MPB galleries and blue staining. Under endemic conditions, MPB will infest weak and dying trees, producing galleries and blue staining. Large-scale coincident mortality will only occur, however, during epidemic outbreak conditions; the combination of MPB galleries, blue stain and coincident mortality is thus a strong indicator of an MPB outbreak. Lodgepole pine forests often establish as even-aged post-fire cohorts, so in addition to the 20 cores from dead lodgepole pine, at least 15 of the oldest trees at each site were cored as close to the ground and to the centre of the tree as possible, in order to determine approximate stand-origin dates (Kipfmueller & Baker, 2000; Kulakowski & Veblen, 2002; Kulakowski *et al.*, 2003; Kulakowski & Jarvis, 2011). Testing in similar forests showed that subjectively selecting the largest trees gave a better estimate of stand age than randomly selecting trees (Kipfmueller & Baker, 1998). Aspect, elevation, slope and stand characteristics (stand structure and composition) were recorded at each site.

Dendroecology and comparative analysis

All tree cores were mounted, sanded, counted and cross-dated in the laboratory (Stokes & Smiley, 1968). For all cores that did not reach the pith, the number of rings to the pith was estimated using Duncan's (1989) method. All cores were measured using a slide-stage micrometer connected to a computer. Cores from live trees for each site were used to create a master ring-width chronology for cross-dating cores from dead trees (assigning calendar dates to each tree ring using iterative correlation) using both the program COFECHA (Holmes, 1983; Grissino-Mayer, 2001) and visual validation. Within COFECHA, a series of segment lengths (segments of successive ring widths) ranging from 60 to 200 years were used in an iterative process of comparisons to find Pearson product–moment correlations between the master chronology and the undated cores from dead trees. A default segment length of 50 years is often used for cross-dating with COFECHA, although tree species with relatively complacent growth (wood growth that does not vary much year to year), like lodgepole pine, may require longer segment lengths to optimize results. Our finished chronologies were further validated through cross-dating with regional chronologies available at the International Tree-Ring Data Bank (National Oceanic and Atmospheric Administration, Washington, DC, USA; http://www.ncdc.noaa.gov/paleo/treering.html) and with in-house chronologies at the Forest

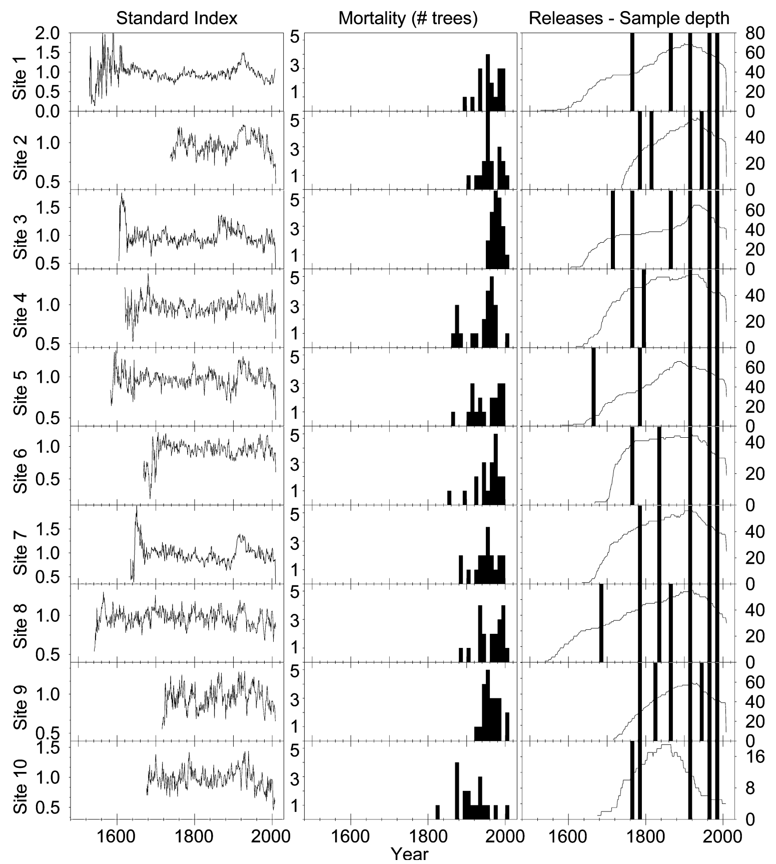

Figure 2 Standard index (standardized ring widths), mortality (last year of growth), and growth releases (interpreted from JOLTS output and cross-site synchrony; histogram represents beginning of release) in 10-year bins at sites 1–10. Sample depth is the number of trees present in the series in each year.

Ecology Lab at Clark University. Trees that could not be cross-dated were not included in subsequent analyses. Each tree-ring series was detrended using a negative exponential function or linear regression and was standardized to a horizontal line (i.e. each ring width was divided by the mean of its series) to permit an averaging that is not biased by faster-growing trees. This method removes rapid initial growth from the record, while preserving low-frequency growth trends throughout the rest of the series. Deviations from average growth rate in these detrended series, which included long periods of increased growth in surviving trees, provided indicators of bark beetle outbreaks (Veblen *et al.*, 1991c).

The initiation of growth releases was quantitatively identified as a twofold increase in ring width (Alfaro *et al.*, 2010; Hrinkevich & Lewis, 2011; Kulakowski & Jarvis, 2011; Kulakowski *et al.*, 2012) using the program JOLTS (Holmes, 1999). Within JOLTS, a moving spline with a 20-year frequency response period was used to attenuate the effects of high-frequency variance in ring widths and provide a conservative method for identifying release initiation periods. A similar method was used by Hrinkevich & Lewis (2011) to eliminate the confounding signal of spruce budworm (*Choristoneura biennis*) when identifying releases associated with MPB. Comparing the patterns of growth across multiple sites highlighted widespread disturbances. In the present study, we therefore designated an outbreak only if the event was recorded in the tree-ring records of at least four sites.

Once historical MPB outbreak dates were reconstructed in our study area, we compared the synchrony of these dates with those reconstructed and documented in previous studies in the USA and Canada. In this context, synchrony refers to the degree to which geographically disjunct MPB outbreak events occurred simultaneously. This comparative analysis was based on published dates of MPB outbreaks. To highlight subcontinental synchrony, we produced a temporal map series where each map represented a period during which a historical MPB outbreak was detected in the present study and/or in previously published sources.

RESULTS

Across all sites, 77% of the sampled cores were successfully cross-dated, yielding samples from 200 dead host trees (18–22 per site), and 376 living host and non-host release trees (4–51 per site). All stands were more than 250 years old, with chronology origin dates ranging from 1566 to 1742. Mortality dates of host trees across all sites ranged from 1818 to 2001, with large clusters of mortality in the 1960s–90s, and smaller clusters in the 1900s–40s and 1940s–60s (Fig. 2). The average return interval (time between consecutive release initiation dates) of disturbance events occurring in at least four sites was 54.6 years, with a standard deviation of 33.5 years and a range of 20–130 years. Following the death of trees killed by MPB, adjacent surviving trees often experienced a release in growth (e.g. Fig. 3). The average release period (time between when release began and when growth returned to the expected negative exponential trend) was 19.5 years, with a standard deviation of 9.5 years and a range of 10–50 years. The interpretation of canopy disturbance dates prior to the 1900s was based only on releases in trees that were growing near old, decomposed lodgepole pine with MPB galleries. Such trees were too decomposed to yield solid core samples.

Sites contained evidence of past MPB outbreak around the 1760s, 1780s, 1820s–1830s, 1860s, 1910s, 1960s and 1980s (Table 1). Sites interpreted as having been affected by an outbreak were characterized by coincident releases in surviving trees and corresponding pulses of mortality of neighbouring host trees with MPB galleries (in the most recent outbreaks *c.* 1900s, 1940s and 1960s–90s). The 1900s event

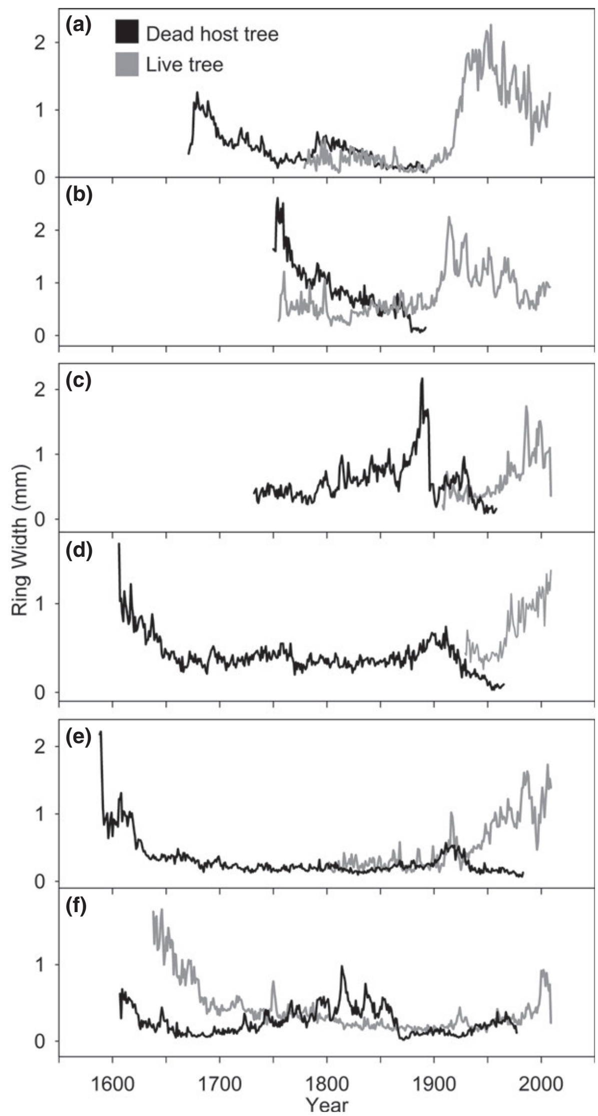

Figure 3 Examples of standardized ring widths in host (black lines) trees killed by *Dendroctonus ponderosae* in *c.* 1900 (a, b), 1960 (c, d), 1980 (e, f), and adjacent live trees (grey lines) experiencing a sudden and sustained increase in growth rate ('release').

was the most intense release in the record, being clearly visible in tree-ring chronologies at all sites (Fig. 2). Sites 2, 8 and 9 had pulses of mortality *c.* 1940s in the tree-ring record, and sites 2 and 9 also had corresponding releases during this period. Although this 1940s event was only confidently recorded at two sites, it was notable that an outbreak was detected during this period in previous studies in the USA and Canada (Tables 1 & 2).

Our comparative analysis indicated that every outbreak detected in previous studies across western North America was temporally synchronous with outbreaks detected in the present study. Furthermore, every outbreak detected in at least four sites in the present study was synchronous with at least one other study in western North America. Our map series (Fig. 4) highlights the subcontinental synchrony of previous outbreaks detected in the present study and in previous studies, and illustrates the similarity in extent of previous synchronous outbreaks to the 2000s–2010s outbreak.

DISCUSSION

Western Colorado has been affected by several extensive outbreaks of MPB over the past several centuries. Furthermore, our continental-scale comparison pointed to broad-scale synchrony of outbreaks across western North America over the past several centuries. These findings put recent and future outbreaks of MPB into a broader ecological context. Our data indicate that broad-scale MPB outbreaks have been important in the development of Rocky Mountain lodgepole pine forests for centuries, although MPB outbreaks in past centuries were probably less contiguous and somewhat less synchronous than more recent ones.

Dynamics of past MPB outbreaks in Colorado

Synchrony of outbreaks among sites in western Colorado varied over time. Recent outbreaks (e.g. 1910s, 1960s and 1980s) were more synchronous across sites than older outbreaks (e.g. 1760s, 1780s and 1820s–1830s). Increased synchrony during more recent outbreaks may have been due to climatic conditions, stand age and/or sample distance decay. Recent climate-driven increases in beetle survival rates (Raffa *et al.*, 2008; Bentz *et al.*, 2009) can underlie severe outbreaks and promote highly synchronous outbreak initiation. Indeed, recent MPB activity was highly synchronous across western Colorado under the warm and dry conditions favourable to

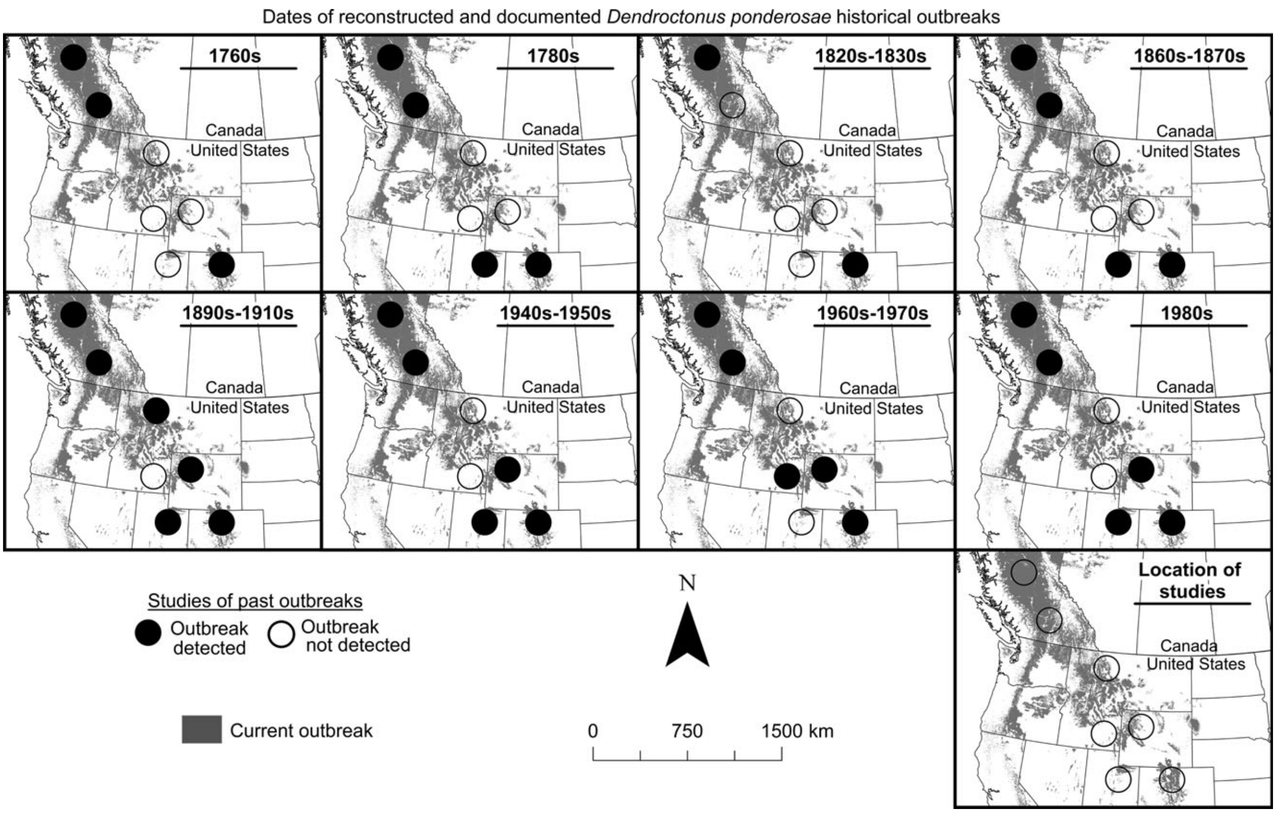

Figure 4 Maps of current and historical *Dendroctonus ponderosae* outbreak in the United States and Canada. Grey shading represents areas affected by the ongoing outbreak, and circles represent generalized areas where historical *Dendroctonus ponderosae* outbreaks have been documented and reconstructed in the current study as well as in previous studies (black circles), and where they have not been detected (empty circles). Empty circles are included in the 'location of studies' map to allow better visualization of the 2000s–2010s outbreak. See Tables 1 and 2 for details of the previous studies and their locations.

MPB (Chapman *et al.*, 2012). Thus, under less favourable climatic conditions, previous regional-scale outbreaks may have been somewhat less synchronous.

Stand age may influence susceptibility to MPB, because young trees and stands in earlier stages of structural development are generally less susceptible to MPB. Taylor & Carroll (2004) found increased susceptibility to MPB with increasing fire cycles up to a threshold of 120 years, when fire cycles became less important in determining susceptibility. Similarly, Kulakowski *et al.* (2012) found that susceptibility to MPB outbreaks generally increased with stand age, with the exception of the most recent outbreak in *c.* 2000–2010s. This decreased influence of pre-outbreak stand conditions may have been due to a threshold in stand development beyond which changes in stand structure no longer affected susceptibility to outbreaks. However, this trend may also be explained by the theoretical expectation that pre-disturbance conditions have a decreasing influence on disturbance susceptibility as the disturbance intensity increases (Turner *et al.*, 1997; Kulakowski & Veblen, 2002). The less synchronous and less widespread outbreaks early in the record of the present study may therefore represent a period of low susceptibility in younger stands and a greater influence of landscape-level variables such as landscape context (forest characteristics of the surrounding landscape, which include host-stand connectivity) and beetle pressure (local abundance of beetle irruptions a few years prior to outbreak) (Simard *et al.*, 2012).

Low synchrony among the oldest outbreaks may also be a methodological artefact. Smaller sample sizes and fewer mortality dates earlier in the record can contribute to less precise estimates of older outbreak dates and less apparent synchrony. The use of tree-rings combined with field evidence has been validated as an effective method for the detection of recent (within the past 50 years) MPB, and has also been shown to be effective for reconstructing older outbreaks (Smith *et al.*, 2012). Nevertheless, early outbreak dates are necessarily approximate, because the lag time between MPB-induced mortality and growth releases in neighbouring trees is variable. Heath & Alfaro (1990) found the lag times between MPB mortality and release of surviving trees (including host and non-host) to range from 1 to 9 years, which is similar to that found in our study. The interpretation of canopy disturbances from release events alone in this study is therefore considered approximate within a decade. Additionally, outbreaks of MPB are not temporally discrete, and can last a decade or more under optimal conditions (Safranyik & Wilson, 2006).

Subcontinental synchrony of MPB outbreaks

The MPB outbreaks reconstructed in this study were synchronous with previously reported outbreaks across western North America (Tables 1 & 2). This finding does not necessarily mean that previous outbreaks were as intense, extensive or contiguous as more recent ones, but it does suggest that previous outbreaks were synchronous across western North America over the past three centuries. Every outbreak reconstructed in the present study (including the less certain 1940s event that was recorded only at two sites) was synchronous with those reconstructed and documented in British Columbia, and all but two were synchronous with reconstructed and documented outbreaks across Colorado, Wyoming, Idaho, Utah and Montana (Tables 1 & 2). The current ongoing outbreak has generally been considered unprecedented in all regards, but our examination of broad-scale and long-term trends of MPB outbreaks suggests that historical outbreaks were likely to be synchronous across large areas and that such synchrony *per se* was not necessarily unprecedented.

Our results extend the current record of historical MPB outbreak in western Colorado by several centuries, and indicate that outbreaks of MPB have been part of the forest dynamics in western Colorado for centuries. Furthermore, these findings, taken together with results from previous studies in the western United States and Canada, strongly suggest the broad-scale synchrony of several MPB outbreaks over the past three centuries across western North America. Future work can help to refine our understanding of past subcontinental-scale MPB outbreaks as well as the long-term climatic drivers of outbreak synchrony.

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